



# KL Divergence as the Geometry of Statistical Evidence

## From Likelihood Ratios to KL-UCB

**Author:** Tang

**Institute:** Tang's Machine Learning Blog

**Date:** 2026-06-19

**Version:** 1.0.0

**Topic:** KL Divergence, Statistical Evidence, KL-UCB

# Contents

|                                                                                |           |
|--------------------------------------------------------------------------------|-----------|
| <b>Chapter 1 Evidence Begins with a Ratio</b>                                  | <b>1</b>  |
| 1.1 Why take the logarithm? . . . . .                                          | 1         |
| 1.2 The sign has a plain meaning . . . . .                                     | 2         |
| <b>Chapter 2 KL Divergence Is Average Log-Evidence</b>                         | <b>3</b>  |
| 2.1 Nats and bits . . . . .                                                    | 3         |
| 2.2 A first numerical reading . . . . .                                        | 4         |
| <b>Chapter 3 Bernoulli KL, Derived from the Data</b>                           | <b>5</b>  |
| 3.1 The likelihood of a binary sample . . . . .                                | 5         |
| 3.2 The sample log-likelihood ratio . . . . .                                  | 5         |
| 3.3 Take the expectation under the true mean . . . . .                         | 6         |
| <b>Chapter 4 Why KL Is Never Negative</b>                                      | <b>7</b>  |
| 4.1 Why it is not an ordinary distance . . . . .                               | 7         |
| <b>Chapter 5 The Local Shape: KL Becomes a Quadratic</b>                       | <b>9</b>  |
| 5.1 Fisher information appears automatically . . . . .                         | 10        |
| 5.2 Hoeffding is the worst-case quadratic shadow . . . . .                     | 10        |
| <b>Chapter 6 The Bernoulli Chernoff Bound, Step by Step</b>                    | <b>12</b> |
| 6.1 Step 1: turn the event into an exponential event . . . . .                 | 12        |
| 6.2 Step 2: apply Markov's inequality . . . . .                                | 12        |
| 6.3 Step 3: use independence . . . . .                                         | 12        |
| 6.4 Step 4: choose the best exponential tilt . . . . .                         | 13        |
| 6.5 Step 5: substitute the optimizer . . . . .                                 | 13        |
| 6.6 KL versus Hoeffding . . . . .                                              | 14        |
| <b>Chapter 7 Confidence Bounds as Inverted Evidence Tests</b>                  | <b>15</b> |
| 7.1 Why the KL radius changes with the mean . . . . .                          | 16        |
| <b>Chapter 8 KL-UCB: Optimism Measured in Evidence</b>                         | <b>17</b> |
| 8.1 The complete decision rule . . . . .                                       | 17        |
| 8.2 What the asymptotic denominator means . . . . .                            | 18        |
| <b>Chapter 9 Information Adds Across Observations</b>                          | <b>19</b> |
| <b>Chapter 10 Information Accounting in an Adaptive Bandit</b>                 | <b>20</b> |
| <b>Chapter 11 A Decision Cannot Contain More Information Than the Data</b>     | <b>22</b> |
| 11.1 The seed of a lower bound . . . . .                                       | 22        |
| <b>Chapter 12 Reproducible Experiment: UCB1, KL-UCB, and Thompson Sampling</b> | <b>24</b> |

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| 12.1 Pseudo-regret . . . . .                                              | 24        |
| 12.2 Where the samples went . . . . .                                     | 25        |
| <b>Chapter 13 Three Ways KL Reappears in Machine Learning</b>             | <b>26</b> |
| 13.1 Variational inference: approximation by information loss . . . . .   | 26        |
| 13.2 Information gain: expected posterior movement . . . . .              | 26        |
| 13.3 Bandit lower bounds and best-arm identification . . . . .            | 26        |
| <b>Chapter 14 Common Misreadings</b>                                      | <b>27</b> |
| 14.1 “KL is the distance between two parameter values” . . . . .          | 27        |
| 14.2 “Large KL means the models are far apart in every sense” . . . . .   | 27        |
| 14.3 “KL is symmetric enough when the parameters are close” . . . . .     | 27        |
| 14.4 “A KL confidence set is automatically valid at every time” . . . . . | 27        |
| <b>Chapter 15 What to Carry Forward</b>                                   | <b>28</b> |
| <b>Appendix A. Formula Sheet</b>                                          | <b>29</b> |
| <b>Appendix B. Notation Table</b>                                         | <b>30</b> |
| <b>Appendix C. Minimal Implementation Notes</b>                           | <b>31</b> |
| <b>Further Reading</b>                                                    | <b>32</b> |
| <b>Appendix D. Full Experiment Script</b>                                 | <b>33</b> |

# Chapter 1 Evidence Begins with a Ratio

Imagine two explanations for the same stream of observations.

- World  $P$ : a recommendation is clicked with probability 0.62.
- World  $Q$ : the same recommendation is clicked with probability 0.50.

A click is possible in both worlds. A non-click is possible in both worlds. One observation therefore does not prove either explanation. It only moves the balance.

For an observed outcome  $x$ , the simplest way to compare the two worlds is

$$\frac{P(x)}{Q(x)}.$$

If the ratio is larger than one,  $x$  is more natural under  $P$ . If it is smaller than one,  $x$  is more natural under  $Q$ .

For a click,

$$\frac{P(X = 1)}{Q(X = 1)} = \frac{0.62}{0.50} = 1.24.$$

For a non-click,

$$\frac{P(X = 0)}{Q(X = 0)} = \frac{0.38}{0.50} = 0.76.$$

A click leans toward  $P$ . A non-click leans toward  $Q$ .

This ratio is called a *likelihood ratio*. The name is less important than the idea:

*Evidence is not the observation by itself. Evidence is how differently the competing explanations predict that observation.*

## Key idea

An event may look impressive and still carry little evidence if both models predict it. An ordinary event may carry strong evidence if one model predicts it and the other nearly rules it out.

## 1.1 Why take the logarithm?

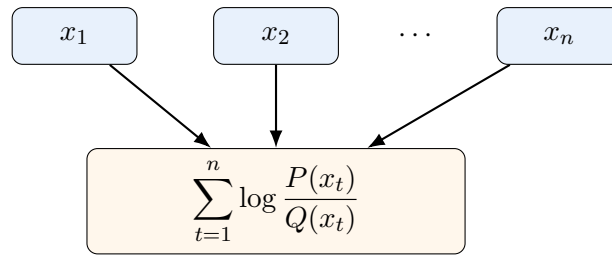
Suppose we observe  $x_1, x_2, \dots, x_n$  independently. The likelihood ratio of the entire sample is

$$\frac{P(x_1, \dots, x_n)}{Q(x_1, \dots, x_n)} = \frac{\prod_{t=1}^n P(x_t)}{\prod_{t=1}^n Q(x_t)} = \prod_{t=1}^n \frac{P(x_t)}{Q(x_t)}.$$

Products are awkward to read. Logs turn them into sums:

$$\begin{aligned} \log \frac{P(x_1, \dots, x_n)}{Q(x_1, \dots, x_n)} &= \log \prod_{t=1}^n \frac{P(x_t)}{Q(x_t)} \\ &= \sum_{t=1}^n \log \frac{P(x_t)}{Q(x_t)}. \end{aligned}$$

One observation contributes one term. The next observation contributes another. Evidence accumulates additively.



cumulative log-evidence for  $P$  against  $Q$

This is why log-likelihood ratios appear everywhere in statistics, information theory, and sequential learning. They are the natural currency in which independent pieces of evidence add.

## 1.2 The sign has a plain meaning

Let

$$L_n = \log \frac{P(X_1, \dots, X_n)}{Q(X_1, \dots, X_n)}.$$

Then

$$L_n > 0 \iff P(X_1, \dots, X_n) > Q(X_1, \dots, X_n),$$

so the observed sample is more likely under  $P$ .

Similarly,

$$L_n < 0 \iff P(X_1, \dots, X_n) < Q(X_1, \dots, X_n),$$

so the observed sample is more likely under  $Q$ .

The number  $L_n$  is random because the data are random. The next question is therefore not whether one particular path rises or falls. It is whether the path has a systematic drift.

## Chapter 2 KL Divergence Is Average Log-Evidence

Suppose  $P$  is the true data-generating distribution. The one-step log-evidence is

$$\log \frac{P(X)}{Q(X)}.$$

Its average under the true world  $P$  is

$$\mathbb{E}_{X \sim P} \left[ \log \frac{P(X)}{Q(X)} \right].$$

This is the Kullback–Leibler divergence:

$$\text{KL}(P \| Q) = \mathbb{E}_{X \sim P} \left[ \log \frac{P(X)}{Q(X)} \right].$$

For a discrete sample space,

$$\text{KL}(P \| Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}.$$

For densities  $p$  and  $q$ ,

$$\text{KL}(P \| Q) = \int p(x) \log \frac{p(x)}{q(x)} dx.$$

The notation  $P \| Q$  is deliberately directional. The expectation is taken under  $P$ . We are asking:

*If  $P$  is true, how much log-evidence does one observation provide against  $Q$ , on average?*

This interpretation, emphasized in classic information-theoretic treatments such as MacKay’s Cambridge lectures and text, is often more useful than calling KL a “distance” [7, 10].

### Key idea

KL divergence is an evidence rate. If  $\text{KL}(P \| Q) = 0.03$ , then one observation contributes about 0.03 nats of evidence for  $P$  against  $Q$  on average. Roughly  $n \text{KL}(P \| Q)$  nats accumulate after  $n$  independent observations.

## 2.1 Nats and bits

Using the natural logarithm gives units called *nats*. Using  $\log_2$  gives bits:

$$\log_2 z = \frac{\log z}{\log 2}.$$

Therefore,

$$\text{KL}_{\text{bits}}(P \| Q) = \frac{\text{KL}_{\text{nats}}(P \| Q)}{\log 2}.$$

The mathematics is unchanged. Only the unit changes.

## 2.2 A first numerical reading

For  $P = \text{Bern}(0.62)$  and  $Q = \text{Bern}(0.50)$ ,

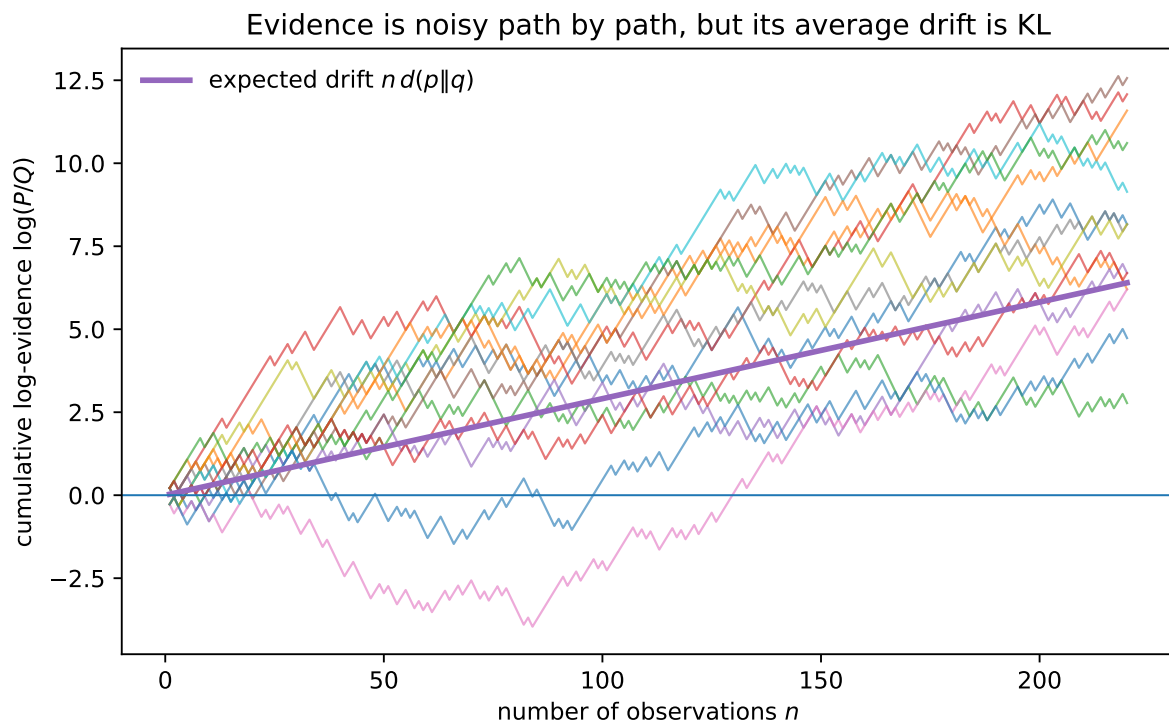
$$\begin{aligned} \text{KL}(P\|Q) &= 0.62 \log \frac{0.62}{0.50} + 0.38 \log \frac{0.38}{0.50} \\ &\approx 0.02908. \end{aligned}$$

After 220 independent observations, the expected cumulative evidence is

$$220 \times 0.02908 \approx 6.40 \text{ nats.}$$

Equivalently, the expected likelihood ratio is not  $e^{6.40}$  because expectation and exponentiation do not commute. What 6.40 means is that the *expected log* likelihood ratio is 6.40.

That distinction matters. Evidence paths are noisy.



**Figure 2.1:** Fourteen independent evidence paths generated under  $P = \text{Bern}(0.62)$  and compared with  $Q = \text{Bern}(0.50)$ . Individual paths wander, while their expected slope is  $\text{KL}(P\|Q)$ .

## Chapter 3 Bernoulli KL, Derived from the Data

The Bernoulli case is the smallest model in which all the ideas are visible.

Let

$$X_t \in \{0, 1\}, \quad X_t \stackrel{\text{iid}}{\sim} \text{Bern}(p).$$

We compare two candidate means  $p$  and  $q$ .

### 3.1 The likelihood of a binary sample

Let

$$S_n = \sum_{t=1}^n X_t$$

be the number of successes. Then the number of failures is  $n - S_n$ .

Under mean  $p$ ,

$$P_p(X_1 = x_1, \dots, X_n = x_n) = \prod_{t=1}^n p^{x_t} (1-p)^{1-x_t}.$$

Collect the powers:

$$\begin{aligned} \prod_{t=1}^n p^{x_t} (1-p)^{1-x_t} &= p^{\sum_{t=1}^n x_t} (1-p)^{\sum_{t=1}^n (1-x_t)} \\ &= p^{S_n} (1-p)^{n-S_n}. \end{aligned}$$

Under mean  $q$ ,

$$P_q(X_1 = x_1, \dots, X_n = x_n) = q^{S_n} (1-q)^{n-S_n}.$$

### 3.2 The sample log-likelihood ratio

Divide the two likelihoods:

$$\begin{aligned} \frac{P_p(X_{1:n})}{P_q(X_{1:n})} &= \frac{p^{S_n} (1-p)^{n-S_n}}{q^{S_n} (1-q)^{n-S_n}} \\ &= \left(\frac{p}{q}\right)^{S_n} \left(\frac{1-p}{1-q}\right)^{n-S_n}. \end{aligned}$$

Take logs:

$$\begin{aligned} L_n(p, q) &= \log \frac{P_p(X_{1:n})}{P_q(X_{1:n})} \\ &= S_n \log \frac{p}{q} + (n - S_n) \log \frac{1-p}{1-q}. \end{aligned}$$

Write  $\hat{p}_n = S_n/n$ :

$$\begin{aligned} L_n(p, q) &= n\hat{p}_n \log \frac{p}{q} + n(1 - \hat{p}_n) \log \frac{1-p}{1-q} \\ &= n \left[ \hat{p}_n \log \frac{p}{q} + (1 - \hat{p}_n) \log \frac{1-p}{1-q} \right]. \end{aligned}$$

### 3.3 Take the expectation under the true mean

Because  $\mathbb{E}_p[\hat{p}_n] = p$ ,

$$\begin{aligned}\mathbb{E}_p[L_n(p, q)] &= n \left[ \mathbb{E}_p[\hat{p}_n] \log \frac{p}{q} + \mathbb{E}_p[1 - \hat{p}_n] \log \frac{1 - p}{1 - q} \right] \\ &= n \left[ p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} \right].\end{aligned}$$

Define

$$\text{kl}(p, q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

Then

$$\mathbb{E}_p[L_n(p, q)] = n \text{kl}(p, q).$$

The formula is not an arbitrary definition placed on top of the problem. It is what remains after we average the log-likelihood ratio generated by the observations.

#### Result

For Bernoulli observations,  $\text{kl}(p, q)$  is the expected one-sample log-evidence for mean  $p$  against mean  $q$ .

## Chapter 4 Why KL Is Never Negative

A single observation can favor the wrong model. A run of observations can also favor the wrong model. KL says that when  $P$  is true, the average log-evidence cannot systematically favor  $Q$ .

The proof uses one elementary inequality:

$$\log u \leq u - 1, \quad u > 0.$$

Equivalently,

$$-\log u \geq 1 - u.$$

Set

$$u(x) = \frac{Q(x)}{P(x)}.$$

Then

$$\begin{aligned} \text{KL}(P\|Q) &= \sum_x P(x) \log \frac{P(x)}{Q(x)} \\ &= \sum_x P(x) \left[ -\log \frac{Q(x)}{P(x)} \right] \\ &\geq \sum_x P(x) \left[ 1 - \frac{Q(x)}{P(x)} \right] \\ &= \sum_x P(x) - \sum_x Q(x) \\ &= 1 - 1 \\ &= 0. \end{aligned}$$

Hence

$$\boxed{\text{KL}(P\|Q) \geq 0.}$$

The inequality  $-\log u \geq 1 - u$  becomes equality only at  $u = 1$ . Therefore, under the usual absolute-continuity conditions,

$$\text{KL}(P\|Q) = 0 \iff P = Q \text{ almost surely.}$$

### Think

Nonnegative does not mean every observation helps. It means helpful observations win on average when the data truly come from  $P$ .

### 4.1 Why it is not an ordinary distance

An ordinary distance satisfies symmetry:

$$d(P, Q) = d(Q, P).$$

KL generally does not:

$$\text{KL}(P\|Q) \neq \text{KL}(Q\|P).$$

There is a simple reason. The first quantity averages under  $P$ ; the second averages under  $Q$ .

For Bernoulli means,

$$\text{kl}(0.10, 0.05) \approx 0.02065,$$

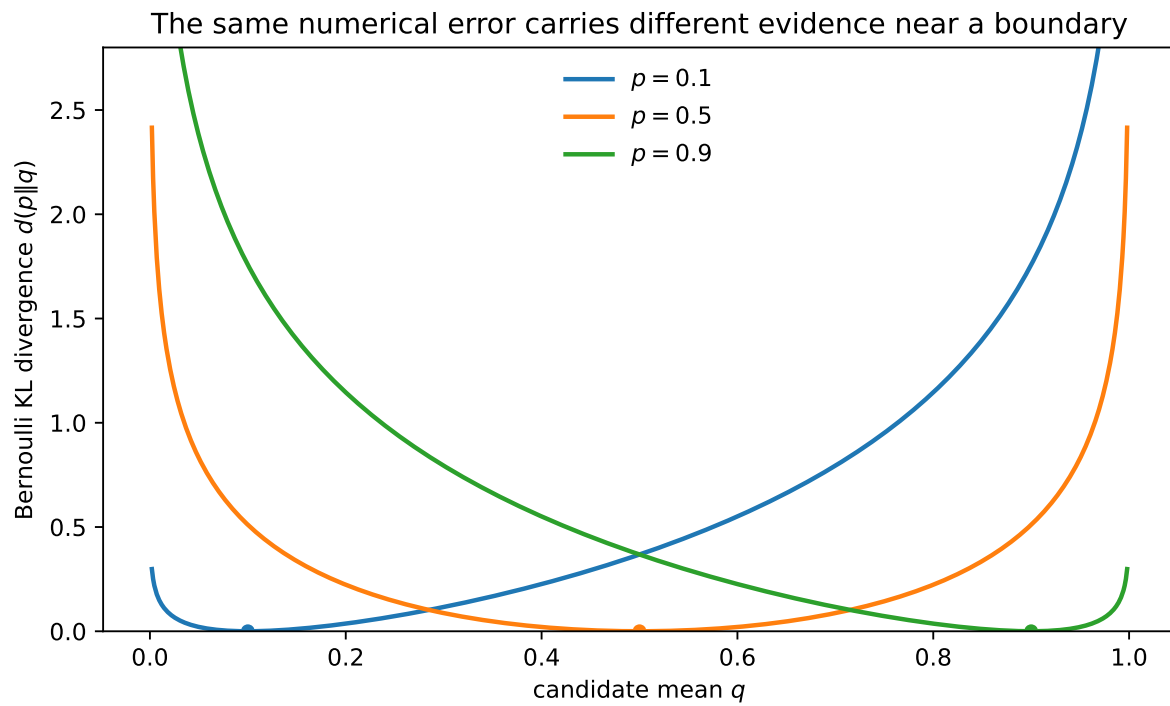
while

$$\text{kl}(0.05, 0.10) \approx 0.01671.$$

The two questions are different:

- If 0.10 is true, how surprising is a model that says 0.05?
- If 0.05 is true, how surprising is a model that says 0.10?

The rare-event structure is different in the two worlds, so the evidence rate is different.



**Figure 4.1:** For a fixed true mean  $p$ , the curve  $q \mapsto \text{kl}(p, q)$  is minimized at  $q = p$ . The geometry changes near the boundaries 0 and 1.

## Chapter 5 The Local Shape: KL Becomes a Quadratic

KL is not a Euclidean distance globally. Near the truth, however, it looks quadratic.

Fix  $p \in (0, 1)$  and write

$$q = p + h,$$

where  $h$  is small. Consider

$$f(q) = \text{kl}(p, q).$$

Differentiate with respect to  $q$ :

$$f'(q) = -\frac{p}{q} + \frac{1-p}{1-q}.$$

At  $q = p$ ,

$$\begin{aligned} f'(p) &= -\frac{p}{p} + \frac{1-p}{1-p} \\ &= -1 + 1 \\ &= 0. \end{aligned}$$

Differentiate again:

$$f''(q) = \frac{p}{q^2} + \frac{1-p}{(1-q)^2}.$$

At  $q = p$ ,

$$\begin{aligned} f''(p) &= \frac{p}{p^2} + \frac{1-p}{(1-p)^2} \\ &= \frac{1}{p} + \frac{1}{1-p} \\ &= \frac{1}{p(1-p)}. \end{aligned}$$

Taylor's formula gives

$$\begin{aligned} \text{kl}(p, p+h) &= f(p) + f'(p)h + \frac{1}{2}f''(p)h^2 + O(h^3) \\ &= 0 + 0 + \frac{h^2}{2p(1-p)} + O(h^3). \end{aligned}$$

Therefore,

$$\text{kl}(p, p+h) = \frac{h^2}{2p(1-p)} + O(h^3).$$

The denominator  $p(1-p)$  is the Bernoulli variance. The same mean error  $h$  is more informative when the observation noise is smaller.

At  $p = 0.50$ ,

$$p(1-p) = 0.25.$$

At  $p = 0.05$ ,

$$p(1-p) = 0.0475.$$

A shift of size 0.02 is therefore much easier to detect near 0.05 than near 0.50.

## 5.1 Fisher information appears automatically

For the Bernoulli family, the Fisher information in one observation is

$$I(p) = \frac{1}{p(1-p)}.$$

The local expansion becomes

$$\text{kl}(p, p+h) = \frac{1}{2}I(p)h^2 + O(h^3).$$

This is the precise sense in which KL supplies a local geometry for statistical models. The curvature is not chosen by hand. It is determined by how quickly the likelihood changes.

### Research lens

This local quadratic structure is the bridge from elementary Bernoulli calculations to exponential families, asymptotic normality, natural gradients, and information geometry. The next chapter will make that bridge explicit.

## 5.2 Hoeffding is the worst-case quadratic shadow

For Bernoulli means, Pinsker's inequality becomes

$$\text{kl}(p, q) \geq 2(p-q)^2.$$

A short calculus proof is possible. Fix  $q$  and define

$$g(p) = \text{kl}(p, q) - 2(p-q)^2.$$

Then

$$g(q) = 0.$$

Also,

$$\begin{aligned} g'(p) &= \log \frac{p(1-q)}{q(1-p)} - 4(p-q), \\ g'(q) &= 0. \end{aligned}$$

Finally,

$$\begin{aligned} g''(p) &= \frac{1}{p} + \frac{1}{1-p} - 4 \\ &= \frac{1}{p(1-p)} - 4 \\ &\geq 0, \end{aligned}$$

where the last step follows from

$$p(1-p) \leq \frac{1}{4}.$$

Thus  $g$  is convex and has a stationary point at  $p = q$ . That point is its minimum, so

$$g(p) \geq g(q) = 0.$$

Hence

$$\boxed{\text{kl}(p, q) \geq 2(p-q)^2.}$$

This inequality explains the relation between KL-based bounds and Hoeffding bounds. Hoeffding replaces

the exact Bernoulli evidence by a universal quadratic lower bound. It is simpler, but it forgets the variance-dependent shape.

## Chapter 6 The Bernoulli Chernoff Bound, Step by Step

We now derive the tail bound that places KL directly inside confidence intervals and bandit algorithms.

Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bern}(p), \quad \hat{p}_n = \frac{1}{n} \sum_{t=1}^n X_t.$$

We want an upper bound on

$$\mathbb{P}_p(\hat{p}_n \geq x), \quad x > p.$$

### 6.1 Step 1: turn the event into an exponential event

For any  $\lambda > 0$ ,

$$\begin{aligned} \hat{p}_n \geq x &\iff \sum_{t=1}^n X_t \geq nx \\ &\iff \exp\left(\lambda \sum_{t=1}^n X_t\right) \geq \exp(\lambda nx). \end{aligned}$$

### 6.2 Step 2: apply Markov's inequality

$$\begin{aligned} \mathbb{P}_p(\hat{p}_n \geq x) &= \mathbb{P}_p\left(\exp\left(\lambda \sum_{t=1}^n X_t\right) \geq e^{\lambda nx}\right) \\ &\leq \frac{\mathbb{E}_p[\exp(\lambda \sum_{t=1}^n X_t)]}{e^{\lambda nx}}. \end{aligned}$$

### 6.3 Step 3: use independence

$$\begin{aligned} \mathbb{E}_p\left[\exp\left(\lambda \sum_{t=1}^n X_t\right)\right] &= \mathbb{E}_p\left[\prod_{t=1}^n e^{\lambda X_t}\right] \\ &= \prod_{t=1}^n \mathbb{E}_p[e^{\lambda X_t}] \\ &= \left(\mathbb{E}_p[e^{\lambda X_1}]\right)^n. \end{aligned}$$

For one Bernoulli variable,

$$\begin{aligned} \mathbb{E}_p[e^{\lambda X_1}] &= (1-p)e^{\lambda \cdot 0} + pe^{\lambda \cdot 1} \\ &= 1 - p + pe^{\lambda}. \end{aligned}$$

Therefore,

$$\mathbb{P}_p(\hat{p}_n \geq x) \leq \exp\left\{n \left[\log(1 - p + pe^{\lambda}) - \lambda x\right]\right\}.$$

## 6.4 Step 4: choose the best exponential tilt

Define

$$\phi(\lambda) = \log(1 - p + pe^\lambda) - \lambda x.$$

Differentiate:

$$\phi'(\lambda) = \frac{pe^\lambda}{1 - p + pe^\lambda} - x.$$

Set  $\phi'(\lambda) = 0$ :

$$\begin{aligned} \frac{pe^\lambda}{1 - p + pe^\lambda} &= x \\ pe^\lambda &= x(1 - p) + xpe^\lambda \\ pe^\lambda(1 - x) &= x(1 - p) \\ e^\lambda &= \frac{x(1 - p)}{p(1 - x)}. \end{aligned}$$

Thus

$$\lambda^* = \log \frac{x(1 - p)}{p(1 - x)}.$$

Because  $x > p$ , we have  $\lambda^* > 0$ .

## 6.5 Step 5: substitute the optimizer

First,

$$\begin{aligned} 1 - p + pe^{\lambda^*} &= 1 - p + p \frac{x(1 - p)}{p(1 - x)} \\ &= 1 - p + \frac{x(1 - p)}{1 - x} \\ &= \frac{(1 - p)(1 - x) + x(1 - p)}{1 - x} \\ &= \frac{1 - p}{1 - x}. \end{aligned}$$

Therefore,

$$\begin{aligned} \phi(\lambda^*) &= \log \frac{1 - p}{1 - x} - x \log \frac{x(1 - p)}{p(1 - x)} \\ &= \log \frac{1 - p}{1 - x} - x \log \frac{x}{p} - x \log \frac{1 - p}{1 - x} \\ &= -x \log \frac{x}{p} + (1 - x) \log \frac{1 - p}{1 - x} \\ &= - \left[ x \log \frac{x}{p} + (1 - x) \log \frac{1 - x}{1 - p} \right] \\ &= -\text{kl}(x, p). \end{aligned}$$

We have proved

$$\mathbb{P}_p(\hat{p}_n \geq x) \leq \exp\{-n \text{kl}(x, p)\}, \quad x > p.$$

Similarly,

$$\mathbb{P}_p(\hat{p}_n \leq x) \leq \exp\{-n \text{kl}(x, p)\}, \quad x < p.$$

**Proof pattern**

The proof pattern is:

rare event  $\rightarrow$  exponential transform  $\rightarrow$  Markov  $\rightarrow$  optimize the exponent  $\rightarrow$  KL divergence.

This pattern reappears in large deviations, sequential tests, confidence sequences, and bandit analysis.

## 6.6 KL versus Hoeffding

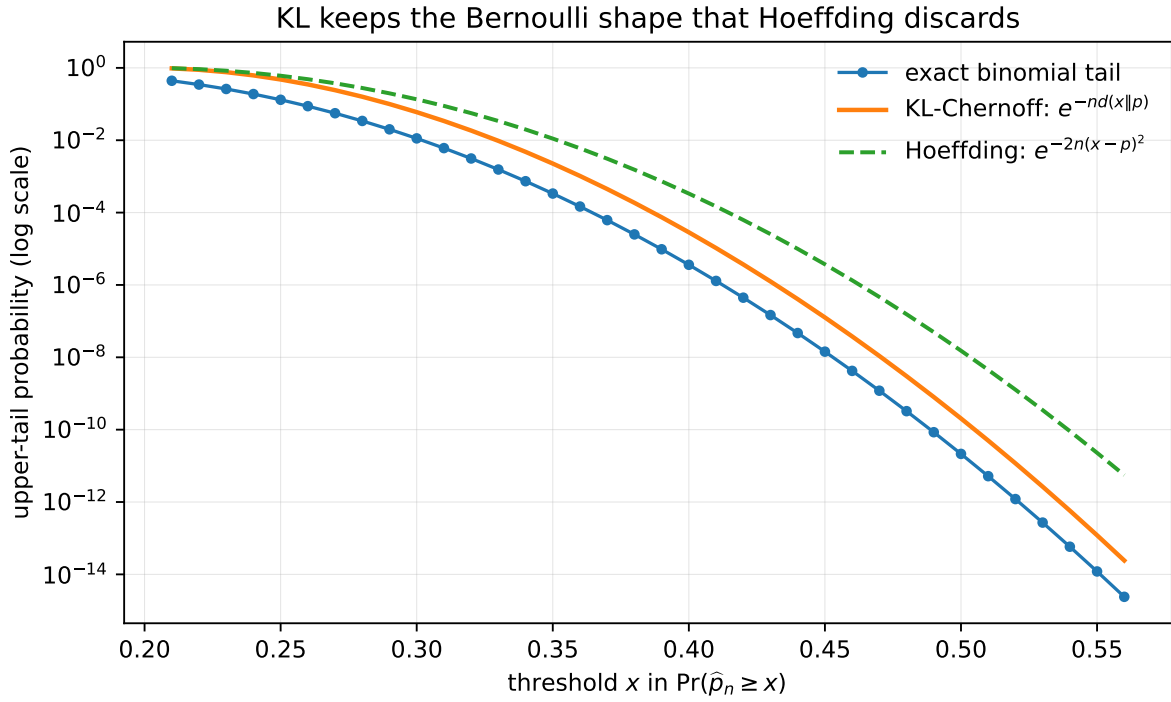
Pinsker gives

$$\text{kl}(x, p) \geq 2(x - p)^2.$$

Therefore,

$$\begin{aligned} \mathbb{P}_p(\hat{p}_n \geq x) &\leq \exp\{-n \text{kl}(x, p)\} \\ &\leq \exp\{-2n(x - p)^2\}. \end{aligned}$$

The KL-Chernoff bound is at least as sharp as the Hoeffding bound in this Bernoulli setting.



**Figure 6.1:** For  $n = 100$  and  $p = 0.20$ , the exact binomial upper tail is compared with its KL-Chernoff and Hoeffding upper bounds. KL preserves more of the distribution's shape.

## Chapter 7 Confidence Bounds as Inverted Evidence Tests

A confidence bound asks which candidate means are still plausible after seeing the data.

Suppose the empirical mean is  $\hat{p}_n$ . A candidate upper mean  $q \geq \hat{p}_n$  becomes harder to believe as

$$n \text{kl}(\hat{p}_n, q)$$

grows. The quantity has a direct reading:

$$\text{sample size} \times \text{evidence per sample}.$$

Given an evidence budget  $\beta > 0$ , define the upper endpoint

$$U_n = \sup \{q \in [\hat{p}_n, 1] : n \text{kl}(\hat{p}_n, q) \leq \beta\}.$$

The set

$$\{q : n \text{kl}(\hat{p}_n, q) \leq \beta\}$$

contains candidate means that the data have not separated strongly enough from the empirical mean.

There is usually no elementary closed form for  $U_n$ . That is not a practical problem. The left side is increasing in  $q$  for  $q \geq \hat{p}_n$ , so binary search finds the endpoint quickly.

```
1 import numpy as np
2
3
4 def bernoulli_kl(p, q):
5     p = np.asarray(p, dtype=float)
6     q = np.clip(np.asarray(q, dtype=float), 1e-14, 1 - 1e-14)
7     p_safe = np.clip(p, 1e-14, 1 - 1e-14)
8
9     first = p_safe * np.log(p_safe / q)
10    second = (1 - p_safe) * np.log((1 - p_safe) / (1 - q))
11    first = np.where(p > 0, first, 0.0)
12    second = np.where(p < 1, second, 0.0)
13    return first + second
14
15
16 def kl_upper_bound(mean, count, beta, steps=28):
17     lo = mean.copy()
18     hi = np.ones_like(mean)
19
20     for _ in range(steps):
21         mid = (lo + hi) / 2
22         feasible = count * bernoulli_kl(mean, mid) <= beta
23         lo = np.where(feasible, mid, lo)
24         hi = np.where(feasible, hi, mid)
25
26     return lo
```

## 7.1 Why the KL radius changes with the mean

Using the local approximation

$$\text{kl}(\hat{p}_n, \hat{p}_n + h) \approx \frac{h^2}{2\hat{p}_n(1 - \hat{p}_n)},$$

the condition

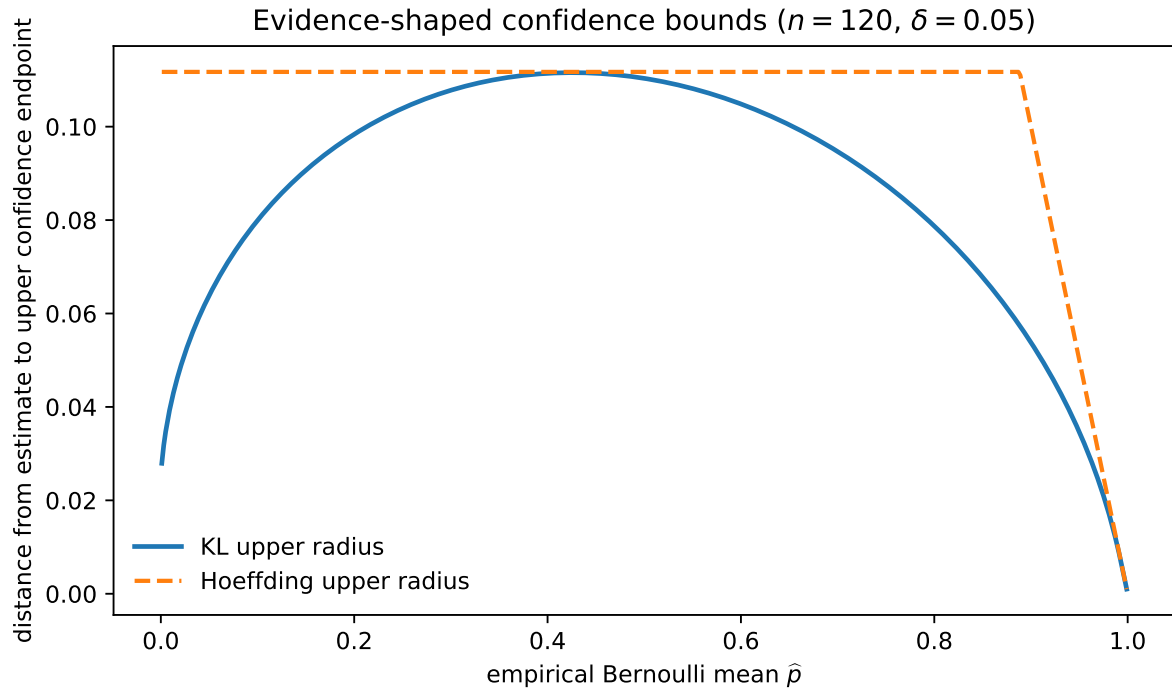
$$n \text{kl}(\hat{p}_n, \hat{p}_n + h) \leq \beta$$

becomes

$$\begin{aligned} n \frac{h^2}{2\hat{p}_n(1 - \hat{p}_n)} &\lesssim \beta \\ h^2 &\lesssim \frac{2\hat{p}_n(1 - \hat{p}_n)\beta}{n} \\ h &\lesssim \sqrt{\frac{2\hat{p}_n(1 - \hat{p}_n)\beta}{n}}. \end{aligned}$$

The radius shrinks near 0 and 1 because Bernoulli variance shrinks there.

Hoeffding uses the worst-case variance  $1/4$  for every mean. It therefore produces an essentially constant radius before clipping at the boundary.



**Figure 7.1:** KL and Hoeffding upper radii with the same evidence budget. KL automatically adapts to the Bernoulli variance.

## Chapter 8 KL-UCB: Optimism Measured in Evidence

UCB chooses the action with the largest plausible mean. The simplest version builds plausibility from a quadratic Hoeffding radius:

$$\hat{\mu}_a(t) + \sqrt{\frac{2 \log t}{N_a(t)}}.$$

KL-UCB keeps the same optimistic principle but replaces the generic radius by a distribution-aware evidence constraint [2].

For Bernoulli rewards, define

$$U_a(t) = \sup \{q \in [\hat{\mu}_a(t), 1] : N_a(t) \text{kl}(\hat{\mu}_a(t), q) \leq f(t)\},$$

where a common exploration function is

$$f(t) = \log t + c \log \log t.$$

Then choose

$$A_t \in \arg \max_a U_a(t).$$

### Key idea

UCB1 asks how far a mean could move under a universal quadratic error bar. KL-UCB asks which larger means remain statistically difficult to distinguish from the observed arm.

### 8.1 The complete decision rule

1. Pull every arm once.
2. For each arm, compute its empirical mean and pull count.
3. Find the largest candidate mean whose KL evidence against the empirical mean is below the exploration budget.
4. Pull the arm with the largest candidate mean.
5. Update only the selected arm.

```
1 for t in range(number_of_arms, horizon):
2     empirical = successes / counts
3
4     log_t = np.log(t + 1.0)
5     beta = log_t + 3.0 * np.log(max(log_t, 1.0))
6
7     index = kl_upper_bound(empirical, counts, beta)
8     chosen = np.argmax(index, axis=1)
9
10    reward = rng.random(number_of_runs) < true_means[chosen]
11    successes[row, chosen] += reward
12    counts[row, chosen] += 1
```

## 8.2 What the asymptotic denominator means

For a suboptimal Bernoulli arm  $a$  with mean  $\mu_a < \mu_*$ , the classical logarithmic scale is

$$\mathbb{E}[N_a(T)] \approx \frac{\log T}{\text{kl}(\mu_a, \mu_*)}.$$

The denominator is the evidence obtained from one pull of arm  $a$  when we compare its true distribution with an alternative in which it could look as good as the best arm.

Large KL:

one pull is informative  $\implies$  few pulls are needed.

Small KL:

the worlds are hard to distinguish  $\implies$  many pulls are unavoidable.

This is the deeper meaning of the Lai–Robbins lower bound and the matching behavior of KL-UCB [2, 8, 9].

## Chapter 9 Information Adds Across Observations

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P$$

under one world and

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} Q$$

under another. The joint laws are  $P^n$  and  $Q^n$ .

The likelihood ratio factorizes:

$$\frac{P^n(X_{1:n})}{Q^n(X_{1:n})} = \prod_{t=1}^n \frac{P(X_t)}{Q(X_t)}.$$

Hence

$$\begin{aligned} \text{KL}(P^n \| Q^n) &= \mathbb{E}_{P^n} \left[ \log \frac{P^n(X_{1:n})}{Q^n(X_{1:n})} \right] \\ &= \mathbb{E}_{P^n} \left[ \sum_{t=1}^n \log \frac{P(X_t)}{Q(X_t)} \right] \\ &= \sum_{t=1}^n \mathbb{E}_P \left[ \log \frac{P(X_t)}{Q(X_t)} \right] \\ &= n \text{KL}(P \| Q). \end{aligned}$$

Thus

$$\boxed{\text{KL}(P^n \| Q^n) = n \text{KL}(P \| Q).}$$

This exact additivity is one reason KL is so central. It turns sample size into an information budget.

## Chapter 10 Information Accounting in an Adaptive Bandit

The data in a bandit are not iid as a single sequence. The selected arm changes with the history. Nevertheless, KL still adds cleanly.

Consider two bandit environments:

$$\nu = (P_1, \dots, P_K), \quad \nu' = (Q_1, \dots, Q_K).$$

Use the same algorithm in both environments. Let the history through time  $T$  be

$$H_T = (A_1, X_1, \dots, A_T, X_T).$$

The algorithm chooses  $A_t$  according to a policy

$$\pi_t(a \mid H_{t-1}).$$

Under environment  $\nu$ , the history density is

$$p_\nu(H_T) = \prod_{t=1}^T \pi_t(A_t \mid H_{t-1}) p_{A_t}(X_t).$$

Under environment  $\nu'$ , it is

$$p_{\nu'}(H_T) = \prod_{t=1}^T \pi_t(A_t \mid H_{t-1}) q_{A_t}(X_t).$$

Divide:

$$\begin{aligned} \frac{p_\nu(H_T)}{p_{\nu'}(H_T)} &= \frac{\prod_{t=1}^T \pi_t(A_t \mid H_{t-1}) p_{A_t}(X_t)}{\prod_{t=1}^T \pi_t(A_t \mid H_{t-1}) q_{A_t}(X_t)} \\ &= \prod_{t=1}^T \frac{p_{A_t}(X_t)}{q_{A_t}(X_t)}. \end{aligned}$$

The policy terms cancel. The algorithm reacts differently because its observed histories differ, but conditional on a given history it uses the same rule in both worlds.

Take logs and expectations under  $\nu$ :

$$\begin{aligned} \text{KL}(\mathbb{P}_\nu^{H_T} \parallel \mathbb{P}_{\nu'}^{H_T}) &= \mathbb{E}_\nu \left[ \sum_{t=1}^T \log \frac{p_{A_t}(X_t)}{q_{A_t}(X_t)} \right] \\ &= \sum_{t=1}^T \mathbb{E}_\nu \left[ \mathbb{E}_\nu \left[ \log \frac{p_{A_t}(X_t)}{q_{A_t}(X_t)} \mid H_{t-1}, A_t \right] \right] \\ &= \sum_{t=1}^T \mathbb{E}_\nu [\text{KL}(P_{A_t} \parallel Q_{A_t})]. \end{aligned}$$

Now use indicators:

$$\text{KL}(P_{A_t} \parallel Q_{A_t}) = \sum_{a=1}^K \mathbf{1}\{A_t = a\} \text{KL}(P_a \parallel Q_a).$$

Therefore,

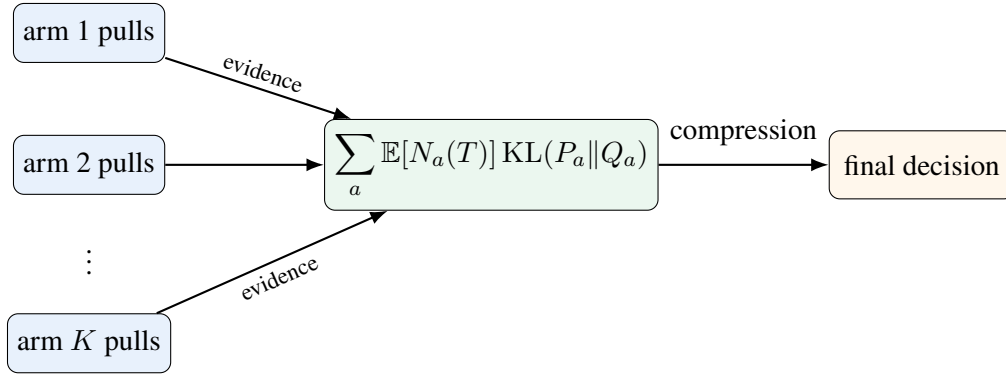
$$\begin{aligned}
 \text{KL}(\mathbb{P}_\nu^{H_T} \parallel \mathbb{P}_{\nu'}^{H_T}) &= \sum_{t=1}^T \sum_{a=1}^K \mathbb{E}_\nu[\mathbf{1}\{A_t = a\}] \text{KL}(P_a \parallel Q_a) \\
 &= \sum_{a=1}^K \mathbb{E}_\nu \left[ \sum_{t=1}^T \mathbf{1}\{A_t = a\} \right] \text{KL}(P_a \parallel Q_a) \\
 &= \sum_{a=1}^K \mathbb{E}_\nu[N_a(T)] \text{KL}(P_a \parallel Q_a).
 \end{aligned}$$

### Proof pattern

Bandit information accounting:

$$\text{total information} = \sum_a \text{expected number of pulls of arm } a \times \text{information per pull}.$$

Adaptivity changes the random pull counts. It does not destroy the accounting identity.



# Chapter 11 A Decision Cannot Contain More Information Than the Data

An algorithm may observe a long transcript and finally output one bit:

$$Z = \mathbf{1}\{\text{recommend arm 1}\}.$$

The output is a compressed version of the history. Compression cannot make two worlds easier to distinguish.

Let  $E$  be any event determined by the data. Write

$$p = P(E), \quad q = Q(E).$$

The binary KL divergence between the event probabilities is

$$\text{kl}(p, q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

The chain rule for the indicator  $Z = \mathbf{1}_E$  gives

$$\begin{aligned} \text{KL}(P\|Q) &= \text{kl}(P(E), Q(E)) \\ &\quad + P(E) \text{KL}(P(\cdot | E)\|Q(\cdot | E)) \\ &\quad + P(E^c) \text{KL}(P(\cdot | E^c)\|Q(\cdot | E^c)). \end{aligned}$$

The last two terms are nonnegative. Therefore,

$$\boxed{\text{KL}(P\|Q) \geq \text{kl}(P(E), Q(E)).}$$

This is a binary form of the data-processing inequality.

## Key idea

The full data may contain subtle evidence. A final yes/no decision can only discard information. Any reliable decision therefore requires enough KL to have accumulated before the compression.

## 11.1 The seed of a lower bound

Suppose an algorithm must identify the best arm in two nearby environments.

Let

$$E = \{\text{algorithm outputs arm 1}\}.$$

In environment  $\nu$ , arm 1 is best, so a  $\delta$ -correct algorithm satisfies

$$\mathbb{P}_\nu(E) \geq 1 - \delta.$$

In environment  $\nu'$ , arm 1 is not best, so

$$\mathbb{P}_{\nu'}(E) \leq \delta.$$

Data processing gives

$$\text{KL}(\mathbb{P}_\nu^{H_T} \| \mathbb{P}_{\nu'}^{H_T}) \geq \text{kl}(1 - \delta, \delta).$$

Combine this with bandit information accounting:

$$\boxed{\sum_{a=1}^K \mathbb{E}_\nu[N_a(T)] \text{KL}(P_a\|Q_a) \geq \text{kl}(1 - \delta, \delta).}$$

For small  $\delta$ ,

$$\text{kl}(1 - \delta, \delta) \asymp \log \frac{1}{\delta}.$$

So reliable identification requires a logarithmic amount of evidence.

This is the central research paradigm behind modern bandit lower bounds and best-arm identification [5].

The next chapter will develop the full change-of-measure method carefully.

## Chapter 12 Reproducible Experiment: UCB1, KL-UCB, and Thompson Sampling

We now run the algorithms in the same four-arm Bernoulli environment:

$$(\mu_1, \mu_2, \mu_3, \mu_4) = (0.03, 0.05, 0.08, 0.12).$$

The horizon is

$$T = 12000,$$

and every curve is averaged over 220 independent runs.

The environment is deliberately near the lower boundary. Bernoulli variance is small there, so a worst-case Hoeffding radius wastes substantial information.

### 12.1 Pseudo-regret

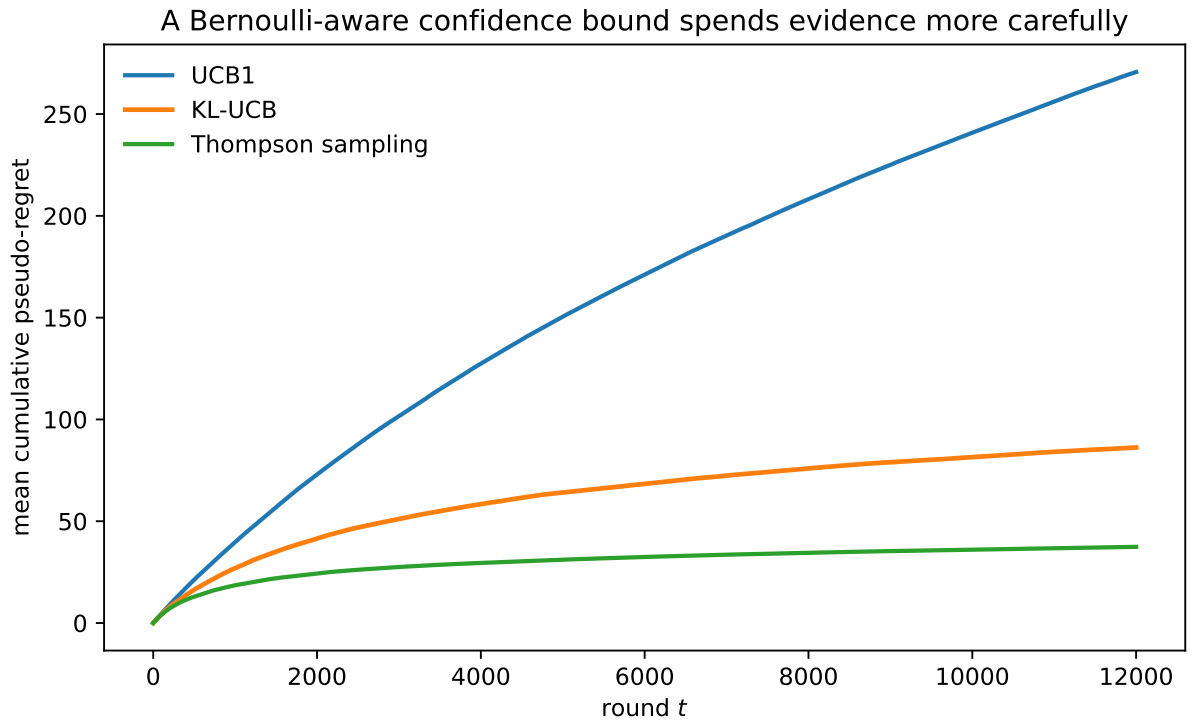
Let

$$\mu_* = 0.12.$$

The cumulative pseudo-regret is

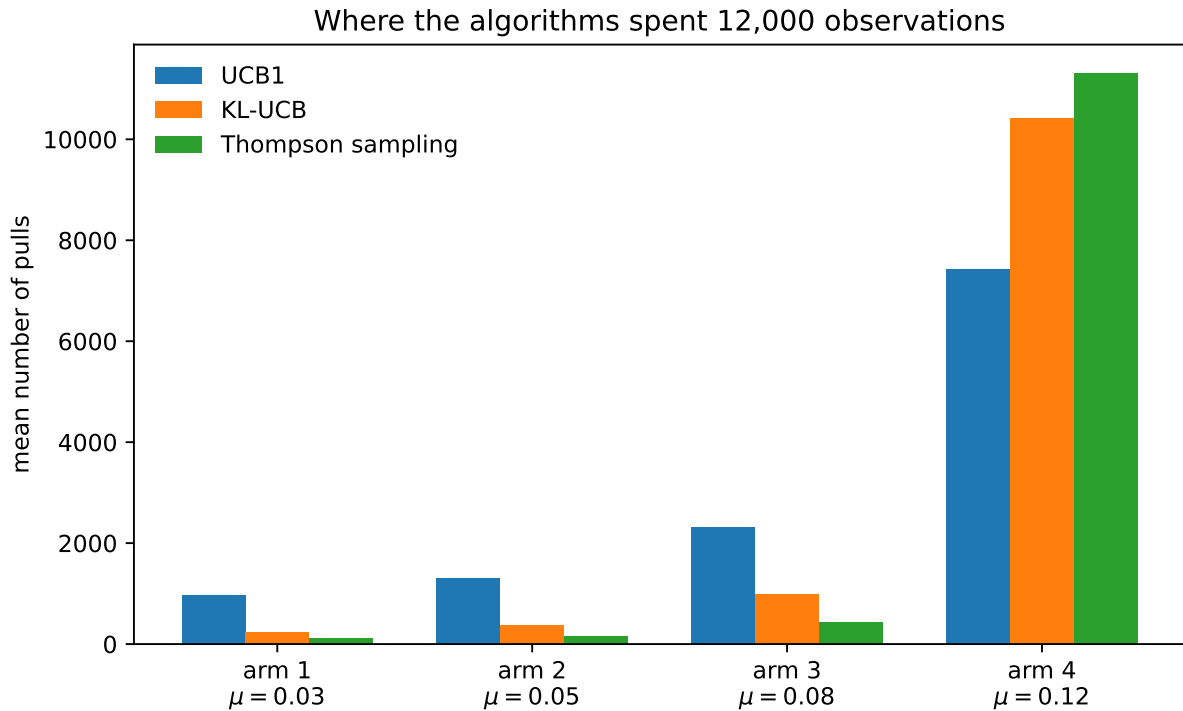
$$R_T = \sum_{t=1}^T (\mu_* - \mu_{A_t}).$$

It depends on the selected arms, not on the realized reward noise. This makes the comparison easier to read.



**Figure 12.1:** Mean cumulative pseudo-regret. KL-UCB uses the Bernoulli evidence geometry and avoids much of the over-exploration of UCB1. Thompson sampling is included as a posterior-sampling reference.

## 12.2 Where the samples went



**Figure 12.2:** Mean pull counts after 12000 rounds. Better evidence accounting means fewer pulls of clearly inferior arms.

**Table 12.1:** Simulation summary. Values are means over 220 runs.

| Algorithm         | Regret | Arm 1  | Arm 2   | Arm 3   | Arm 4    |
|-------------------|--------|--------|---------|---------|----------|
| UCB1              | 270.64 | 967.43 | 1303.31 | 2308.53 | 7420.74  |
| KL-UCB            | 86.20  | 227.61 | 372.99  | 990.07  | 10409.33 |
| Thompson sampling | 37.42  | 103.01 | 156.88  | 429.17  | 11310.94 |

The experiment is not a theorem. Different instances and horizons can change the numerical ranking. The durable lesson is structural:

a confidence rule that matches the reward model

can use observations more efficiently than

a confidence rule built from a worst-case surrogate.

## Chapter 13 Three Ways KL Reappears in Machine Learning

### 13.1 Variational inference: approximation by information loss

Suppose  $p(z \mid x)$  is a difficult posterior and  $q(z)$  is a tractable approximation. Variational inference often minimizes

$$\text{KL}(q(z) \parallel p(z \mid x)).$$

The same divergence now measures the average log mismatch when samples come from the approximation  $q$ . The direction matters: reversing the arguments changes which parts of the posterior are expensive to miss.

This probabilistic-learning view is part of the Cambridge tradition associated with MacKay and Ghahramani: probability models are not decorative wrappers around algorithms; they determine the objective and the geometry [3, 10].

### 13.2 Information gain: expected posterior movement

Mutual information is an expected KL divergence:

$$I(\Theta; Y) = \mathbb{E}_Y [\text{KL}(P(\Theta \mid Y) \parallel P(\Theta))].$$

It measures how far the posterior is expected to move after seeing  $Y$ .

This idea drives information-directed sampling and information-based Bayesian optimization. In work connected to ETH Zurich, information gain controls regret in Gaussian-process and heteroscedastic bandits [6, 13]. Oxford work on information-theoretic Bayesian optimization similarly chooses evaluations for their expected reduction of uncertainty about the optimum [11].

### 13.3 Bandit lower bounds and best-arm identification

In best-arm identification, the learner must collect enough evidence to rule out every alternative world in which another arm is best. Each arm contributes evidence at a rate determined by KL. The sample allocation problem is therefore an information allocation problem.

This viewpoint is central to the lower bounds and algorithms of Kaufmann, Cappe, and Garivier [5], and it continues in recent work on minimal-regret best-arm identification by Yang, Tan, and Tianyuan Jin [14]. Their Double KL-UCB construction makes the connection especially explicit: confidence bounds are not only a technical device; they are the mechanism by which regret and identification evidence are jointly managed.

## Chapter 14 Common Misreadings

### 14.1 “KL is the distance between two parameter values”

Not exactly. KL compares probability laws. Parameters enter only through the distributions they define. For Bernoulli models,

$$p \mapsto \text{Bern}(p),$$

so writing  $\text{kl}(p, q)$  is harmless shorthand. In a different model family, the same numerical parameter gap may generate a very different KL divergence.

### 14.2 “Large KL means the models are far apart in every sense”

KL is tailored to statistical discrimination. It need not agree with Euclidean distance, Wasserstein distance, or total variation. Different divergences answer different questions.

### 14.3 “KL is symmetric enough when the parameters are close”

Locally, both directions share the same leading quadratic term:

$$\text{KL}(P_\theta \| P_{\theta+h}) = \frac{1}{2} h^\top I(\theta) h + o(\|h\|^2).$$

But outside a small neighborhood, direction can matter greatly.

### 14.4 “A KL confidence set is automatically valid at every time”

No. The shape of a confidence set and its time validity are separate issues. The martingale chapter explained why repeated monitoring requires time-uniform control. A KL-shaped boundary can be fixed-time or anytime-valid depending on how the threshold is calibrated.

## Chapter 15 What to Carry Forward

A likelihood ratio compares how well two worlds explain the same observation.

A log-likelihood ratio makes evidence additive.

KL divergence is the expected log-likelihood ratio under the true world.

For Bernoulli observations,

$$\text{kl}(p, q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

Near the truth,

$$\text{kl}(p, p + h) \approx \frac{h^2}{2p(1 - p)}.$$

For independent observations,

$$\text{KL}(P^n \| Q^n) = n \text{KL}(P \| Q).$$

For an adaptive bandit,

$$\text{KL}(\mathbb{P}_\nu^{H_T} \| \mathbb{P}_{\nu'}^{H_T}) = \sum_a \mathbb{E}_\nu[N_a(T)] \text{KL}(P_a \| Q_a).$$

For any final decision event  $E$ ,

$$\text{KL}(P \| Q) \geq \text{kl}(P(E), Q(E)).$$

Together, these identities say:

samples create evidence,   actions allocate evidence,   decisions consume evidence.

That is the role of KL divergence in bandit theory.

## Appendix A. Formula Sheet

Core formulas.

| Object                      | Formula                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|
| Likelihood ratio            | $P(x)/Q(x)$                                                                                                    |
| Log-likelihood ratio        | $\log(P(x)/Q(x))$                                                                                              |
| KL divergence               | $\text{KL}(P\ Q) = \mathbb{E}_P[\log(P(X)/Q(X))]$                                                              |
| Bernoulli KL                | $\text{kl}(p, q) = p \log(p/q) + (1 - p) \log((1 - p)/(1 - q))$                                                |
| Nonnegativity               | $\text{KL}(P\ Q) \geq 0$                                                                                       |
| Local Bernoulli expansion   | $\text{kl}(p, p + h) = h^2/[2p(1 - p)] + O(h^3)$                                                               |
| Bernoulli Pinsker           | $\text{kl}(p, q) \geq 2(p - q)^2$                                                                              |
| KL-Chernoff upper tail      | $\mathbb{P}_p(\hat{p}_n \geq x) \leq e^{-n \text{kl}(x, p)}, x > p$                                            |
| KL upper endpoint           | $\sup\{q \geq \hat{p} : n \text{kl}(\hat{p}, q) \leq \beta\}$                                                  |
| Product additivity          | $\text{KL}(P^n\ Q^n) = n \text{KL}(P\ Q)$                                                                      |
| Bandit information identity | $\text{KL}(\mathbb{P}_\nu^{H_T}\ \mathbb{P}_{\nu'}^{H_T}) = \sum_a \mathbb{E}_\nu[N_a(T)] \text{KL}(P_a\ Q_a)$ |
| Binary data processing      | $\text{KL}(P\ Q) \geq \text{kl}(P(E), Q(E))$                                                                   |
| Mutual information          | $I(\Theta; Y) = \mathbb{E}_Y[\text{KL}(P(\Theta   Y)\ P(\Theta))]$                                             |

## Appendix B. Notation Table

| Notation.         |                                                                             |
|-------------------|-----------------------------------------------------------------------------|
| Symbol            | Meaning                                                                     |
| $P, Q$            | competing probability distributions                                         |
| $p, q$            | Bernoulli means or probability mass/density functions, according to context |
| $X_t$             | observation at time $t$                                                     |
| $S_n$             | number of Bernoulli successes in $n$ observations                           |
| $\hat{p}_n$       | empirical Bernoulli mean $S_n/n$                                            |
| $L_n(P, Q)$       | cumulative log-likelihood ratio for $P$ against $Q$                         |
| $\text{KL}(P\ Q)$ | expected log-evidence under $P$ against $Q$                                 |
| $\text{kl}(p, q)$ | binary/Bernoulli KL divergence                                              |
| $A_t$             | arm chosen at time $t$                                                      |
| $N_a(T)$          | number of pulls of arm $a$ through time $T$                                 |
| $H_T$             | complete bandit history through time $T$                                    |
| $U_a(t)$          | upper confidence index of arm $a$                                           |
| $\delta$          | target error probability                                                    |
| $\beta$           | evidence threshold used in a confidence set                                 |

## Appendix C. Minimal Implementation Notes

The supplied Python script produces every figure and table in this note. The two numerically important details are:

1. Clip candidate probabilities away from exactly 0 and 1 before taking logs.
2. Preserve the correct limiting convention  $0 \log 0 = 0$  when an empirical mean is exactly 0 or 1.

The KL-UCB endpoint is monotone and can be found by binary search. Nineteen to thirty iterations are usually more than enough for double-precision simulation.

## Further Reading

MacKay's *Information Theory, Inference, and Learning Algorithms* is an unusually clear route from likelihood ratios to modern probabilistic learning. Lattimore and Szepesvari place KL directly inside the language of bandits. Garivier and Cappé develop KL-UCB. Kaufmann, Cappé, and Garivier show how KL becomes an information budget for best-arm identification. For information-directed algorithms, the work of Russo and Van Roy and the ETH line around Krause provide the natural continuation.

## Appendix D. Full Experiment Script

```
1  """Reproducible experiments for
2  'KL Divergence as the Geometry of Statistical Evidence'.
3
4  The script creates:
5  - bernoulli_kl_geometry.pdf/png
6  - evidence_accumulation.pdf/png
7  - tail_bounds_comparison.pdf/png
8  - confidence_radii.pdf/png
9  - kl_ucb_regret.pdf/png
10 - kl_ucb_pull_counts.pdf/png
11 - kl_divergence_results.csv
12
13 All simulations use fixed random seeds.
14 """
15
16 from __future__ import annotations
17
18 from dataclasses import dataclass
19 from pathlib import Path
20 import math
21
22 import matplotlib.pyplot as plt
23 import numpy as np
24 import pandas as pd
25 from scipy.stats import binom
26
27
28 OUT = Path(__file__).resolve().parent
29 SEED = 20260619
30
31
32 @dataclass(frozen=True)
33 class Config:
34     evidence_p: float = 0.62
35     evidence_q: float = 0.50
36     evidence_horizon: int = 220
37     evidence_paths: int = 14
38
39     tail_p: float = 0.20
40     tail_n: int = 100
41
42     confidence_n: int = 120
43     confidence_delta: float = 0.05
44
45     bandit_means: tuple[float, ...] = (0.03, 0.05, 0.08, 0.12)
```

```

46     bandit_horizon: int = 12000
47     bandit_runs: int = 220
48     kl_binary_steps: int = 19
49
50
51 def bernoulli_kl(p: np.ndarray | float, q: np.ndarray | float) -> np.ndarray:
52     """Binary relative entropy  $d(p || q)$ , with stable boundary handling."""
53     p_arr = np.asarray(p, dtype=float)
54     q_arr = np.asarray(q, dtype=float)
55     q_safe = np.clip(q_arr, 1e-14, 1.0 - 1e-14)
56
57     p_safe = np.clip(p_arr, 1e-14, 1.0 - 1e-14)
58     first_raw = p_safe * np.log(p_safe / q_safe)
59     second_raw = (1.0 - p_safe) * np.log((1.0 - p_safe) / (1.0 - q_safe))
60     first = np.where(p_arr > 0.0, first_raw, 0.0)
61     second = np.where(p_arr < 1.0, second_raw, 0.0)
62     return first + second
63
64
65 def kl_upper_bound(
66     empirical_mean: np.ndarray,
67     counts: np.ndarray,
68     beta: float,
69     steps: int = 28,
70 ) -> np.ndarray:
71     """Largest  $q$  in  $[empirical\_mean, 1]$  with  $N d(mean || q) \leq \beta$ ."""
72     lo = empirical_mean.copy()
73     hi = np.ones_like(empirical_mean)
74     for _ in range(steps):
75         mid = (lo + hi) / 2.0
76         feasible = counts * bernoulli_kl(empirical_mean, mid) <= beta
77         lo = np.where(feasible, mid, lo)
78         hi = np.where(feasible, hi, mid)
79     return lo
80
81
82 def plot_bernoulli_kl_geometry() -> pd.DataFrame:
83     q = np.linspace(0.002, 0.998, 800)
84     reference_ps = (0.10, 0.50, 0.90)
85
86     fig, ax = plt.subplots(figsize=(7.2, 4.45))
87     for p in reference_ps:
88         ax.plot(q, bernoulli_kl(p, q), linewidth=1.9, label=fr"$p={p:.1f}$")
89         ax.scatter([p], [0.0], s=24)
90     ax.set_xlabel("candidate mean  $q$ ")
91     ax.set_ylabel(r"Bernoulli KL divergence  $d(p \backslash \text{Vert } q)$ ")
92     ax.set_ylim(0.0, 2.8)
93     ax.set_title("The same numerical error carries different evidence near a boundary")

```

```

94     ax.legend(frameon=False)
95     fig.tight_layout()
96     fig.savefig(OUT / "bernoulli_kl_geometry.pdf", bbox_inches="tight")
97     fig.savefig(OUT / "bernoulli_kl_geometry.png", dpi=220, bbox_inches="tight")
98     plt.close(fig)
99
100     rows = []
101     for p in reference_ps:
102         for offset in (-0.05, 0.05):
103             candidate = p + offset
104             if 0.0 < candidate < 1.0:
105                 rows.append(
106                     {
107                         "experiment": "geometry",
108                         "quantity": f"d({p:.2f}||{candidate:.2f})",
109                         "value": float(bernoulli_kl(p, candidate)),
110                     }
111                 )
112     return pd.DataFrame(rows)
113
114
115 def plot_evidence_accumulation(rng: np.random.Generator, cfg: Config) -> pd.DataFrame:
116     p = cfg.evidence_p
117     q = cfg.evidence_q
118     n = cfg.evidence_horizon
119     x = rng.random((cfg.evidence_paths, n)) < p
120     increments = np.where(x, math.log(p / q), math.log((1.0 - p) / (1.0 - q)))
121     paths = np.cumsum(increments, axis=1)
122     time = np.arange(1, n + 1)
123     expected = time * float(bernoulli_kl(p, q))
124
125     fig, ax = plt.subplots(figsize=(7.2, 4.55))
126     for path in paths:
127         ax.plot(time, path, linewidth=0.95, alpha=0.62)
128     ax.plot(time, expected, linewidth=2.4, label=r"expected drift  $\mathbb{E}[\log(P/Q)]$ ")
129     ax.axhline(0.0, linewidth=0.85)
130     ax.set_xlabel("number of observations  $n$ ")
131     ax.set_ylabel(r"cumulative log-evidence  $\log(P/Q)$ ")
132     ax.set_title("Evidence is noisy path by path, but its average drift is KL")
133     ax.legend(frameon=False, loc="upper left")
134     fig.tight_layout()
135     fig.savefig(OUT / "evidence_accumulation.pdf", bbox_inches="tight")
136     fig.savefig(OUT / "evidence_accumulation.png", dpi=220, bbox_inches="tight")
137     plt.close(fig)
138
139     audit_x = rng.random((20000, n)) < p
140     audit_increments = np.where(
141         audit_x,

```

```

142     math.log(p / q),
143     math.log((1.0 - p) / (1.0 - q)),
144 )
145 audit_final = audit_increments.sum(axis=1)
146
147 return pd.DataFrame(
148     [
149         {
150             "experiment": "evidence",
151             "quantity": "KL per observation",
152             "value": float(bernoulli_kl(p, q)),
153         },
154         {
155             "experiment": "evidence",
156             "quantity": "theoretical expected final log evidence",
157             "value": float(expected[-1]),
158         },
159         {
160             "experiment": "evidence",
161             "quantity": "simulated mean final log evidence (20000 paths)",
162             "value": float(audit_final.mean()),
163         },
164     ]
165 )
166
167
168 def plot_tail_bounds(cfg: Config) -> pd.DataFrame:
169     p = cfg.tail_p
170     n = cfg.tail_n
171     counts = np.arange(math.floor(n * p) + 1, math.floor(0.56 * n) + 1)
172     x = counts / n
173
174     exact = binom.sf(counts - 1, n, p)
175     kl_bound = np.exp(-n * bernoulli_kl(x, p))
176     hoeffding = np.exp(-2.0 * n * (x - p) ** 2)
177
178     fig, ax = plt.subplots(figsize=(7.2, 4.45))
179     ax.semilogy(x, exact, marker="o", markersize=3.5, linewidth=1.4, label="exact binomial
180     tail")
181     ax.semilogy(x, kl_bound, linewidth=1.9, label=r"KL-Chernoff:  $e^{-n d(x \parallel p)}$ ")
182     ax.semilogy(x, hoeffding, linewidth=1.7, linestyle="--", label=r"Hoeffding:  $e^{-2n(x-p)^2}$ ")
183     ax.set_xlabel(r"threshold  $x$  in  $\Pr(\widehat{p}_n \geq x)$ ")
184     ax.set_ylabel("upper-tail probability (log scale)")
185     ax.set_title("KL keeps the Bernoulli shape that Hoeffding discards")
186     ax.legend(frameon=False)
187     ax.grid(True, which="both", linewidth=0.35, alpha=0.35)
188     fig.tight_layout()

```

```

188 fig.savefig(OUT / "tail_bounds_comparison.pdf", bbox_inches="tight")
189 fig.savefig(OUT / "tail_bounds_comparison.png", dpi=220, bbox_inches="tight")
190 plt.close(fig)
191
192 selected = [0, len(x) // 3, 2 * len(x) // 3, len(x) - 1]
193 rows = []
194 for j in selected:
195     rows.extend(
196         [
197             {
198                 "experiment": "tail_bound",
199                 "quantity": f"exact tail at x={x[j]:.2f}",
200                 "value": float(exact[j]),
201             },
202             {
203                 "experiment": "tail_bound",
204                 "quantity": f"KL bound at x={x[j]:.2f}",
205                 "value": float(kl_bound[j]),
206             },
207             {
208                 "experiment": "tail_bound",
209                 "quantity": f"Hoeffding bound at x={x[j]:.2f}",
210                 "value": float(hoeffding[j]),
211             },
212         ]
213     )
214 return pd.DataFrame(rows)
215
216
217 def plot_confidence_radii(cfg: Config) -> pd.DataFrame:
218     empirical = np.linspace(0.001, 0.999, 500)
219     counts = np.full_like(empirical, cfg.confidence_n, dtype=float)
220     beta = math.log(1.0 / cfg.confidence_delta)
221     kl_upper = kl_upper_bound(empirical, counts, beta, steps=32)
222     kl_radius = kl_upper - empirical
223     hoeffding_radius = np.minimum(
224         1.0 - empirical,
225         math.sqrt(beta / (2.0 * cfg.confidence_n)) * np.ones_like(empirical),
226     )
227
228     fig, ax = plt.subplots(figsize=(7.2, 4.4))
229     ax.plot(empirical, kl_radius, linewidth=2.0, label="KL upper radius")
230     ax.plot(empirical, hoeffding_radius, linewidth=1.8, linestyle="--", label="Hoeffding
231         upper radius")
232     ax.set_xlabel(r"empirical Bernoulli mean  $\widehat{p}$ ")
233     ax.set_ylabel("distance from estimate to upper confidence endpoint")
234     ax.set_title(fr"Evidence-shaped confidence bounds ( $n=\{cfg.confidence\_n\}$ ,  $\delta=\{cfg.confidence\_delta\}$ )")

```

```

234     ax.legend(frameon=False)
235     fig.tight_layout()
236     fig.savefig(OUT / "confidence_radii.pdf", bbox_inches="tight")
237     fig.savefig(OUT / "confidence_radii.png", dpi=220, bbox_inches="tight")
238     plt.close(fig)
239
240     rows = []
241     for p_hat in (0.02, 0.10, 0.50, 0.90):
242         j = int(np.argmin(np.abs(empirical - p_hat)))
243         rows.extend(
244             [
245                 {
246                     "experiment": "confidence_radius",
247                     "quantity": f"KL radius at mean={p_hat:.2f}",
248                     "value": float(kl_radius[j]),
249                 },
250                 {
251                     "experiment": "confidence_radius",
252                     "quantity": f"Hoeffding radius at mean={p_hat:.2f}",
253                     "value": float(hoeffding_radius[j]),
254                 },
255             ]
256         )
257     return pd.DataFrame(rows)
258
259
260 def _initialize_bandit(
261     rng: np.random.Generator,
262     means: np.ndarray,
263     runs: int,
264 ) -> tuple[np.ndarray, np.ndarray, np.ndarray, np.ndarray]:
265     k = len(means)
266     successes = np.zeros((runs, k), dtype=np.int32)
267     counts = np.zeros((runs, k), dtype=np.int32)
268     cumulative_regret = np.zeros(runs, dtype=float)
269     curve = np.zeros(1, dtype=float)
270
271     for arm in range(k):
272         reward = rng.random(runs) < means[arm]
273         successes[:, arm] += reward
274         counts[:, arm] += 1
275         cumulative_regret += means.max() - means[arm]
276
277     return successes, counts, cumulative_regret, curve
278
279
280 def simulate_ucb(
281     rng: np.random.Generator,

```

```

282     means: np.ndarray,
283     horizon: int,
284     runs: int,
285 ) -> tuple[np.ndarray, np.ndarray]:
286     k = len(means)
287     successes, counts, cumulative_regret, _ = _initialize_bandit(rng, means, runs)
288     curve = np.zeros(horizon + 1, dtype=float)
289
290     running = 0.0
291     for arm in range(k):
292         running += means.max() - means[arm]
293         curve[arm + 1] = running
294     curve[k] = cumulative_regret.mean()
295
296     row = np.arange(runs)
297     for t in range(k, horizon):
298         empirical = successes / counts
299         index = empirical + np.sqrt(2.0 * math.log(t + 1.0) / counts)
300         chosen = np.argmax(index, axis=1)
301         reward = rng.random(runs) < means[chosen]
302         successes[row, chosen] += reward
303         counts[row, chosen] += 1
304         cumulative_regret += means.max() - means[chosen]
305         curve[t + 1] = cumulative_regret.mean()
306     return curve, counts.mean(axis=0)
307
308
309 def simulate_kl_ucb(
310     rng: np.random.Generator,
311     means: np.ndarray,
312     horizon: int,
313     runs: int,
314     binary_steps: int,
315 ) -> tuple[np.ndarray, np.ndarray]:
316     k = len(means)
317     successes, counts, cumulative_regret, _ = _initialize_bandit(rng, means, runs)
318     curve = np.zeros(horizon + 1, dtype=float)
319
320     running = 0.0
321     for arm in range(k):
322         running += means.max() - means[arm]
323         curve[arm + 1] = running
324     curve[k] = cumulative_regret.mean()
325
326     row = np.arange(runs)
327     for t in range(k, horizon):
328         empirical = successes / counts
329         log_t = math.log(t + 1.0)

```

```

330     beta = log_t + 3.0 * math.log(max(log_t, 1.0))
331     index = kl_upper_bound(empirical, counts.astype(float), beta, steps=binary_steps)
332     chosen = np.argmax(index, axis=1)
333     reward = rng.random(runs) < means[chosen]
334     successes[row, chosen] += reward
335     counts[row, chosen] += 1
336     cumulative_regret += means.max() - means[chosen]
337     curve[t + 1] = cumulative_regret.mean()
338     return curve, counts.mean(axis=0)
339
340
341 def simulate_thompson(
342     rng: np.random.Generator,
343     means: np.ndarray,
344     horizon: int,
345     runs: int,
346 ) -> tuple[np.ndarray, np.ndarray]:
347     k = len(means)
348     successes, counts, cumulative_regret, _ = _initialize_bandit(rng, means, runs)
349     curve = np.zeros(horizon + 1, dtype=float)
350
351     running = 0.0
352     for arm in range(k):
353         running += means.max() - means[arm]
354         curve[arm + 1] = running
355     curve[k] = cumulative_regret.mean()
356
357     row = np.arange(runs)
358     for t in range(k, horizon):
359         samples = rng.beta(successes + 1.0, counts - successes + 1.0)
360         chosen = np.argmax(samples, axis=1)
361         reward = rng.random(runs) < means[chosen]
362         successes[row, chosen] += reward
363         counts[row, chosen] += 1
364         cumulative_regret += means.max() - means[chosen]
365         curve[t + 1] = cumulative_regret.mean()
366     return curve, counts.mean(axis=0)
367
368
369 def plot_bandit_experiment(cfg: Config) -> pd.DataFrame:
370     means = np.asarray(cfg.bandit_means, dtype=float)
371
372     ucb_curve, ucb_counts = simulate_ucb(
373         np.random.default_rng(SEED + 101), means, cfg.bandit_horizon, cfg.bandit_runs
374     )
375     kl_curve, kl_counts = simulate_kl_ucb(
376         np.random.default_rng(SEED + 202),
377         means,

```

```

378         cfg.bandit_horizon,
379         cfg.bandit_runs,
380         cfg.kl_binary_steps,
381     )
382     ts_curve, ts_counts = simulate_thompson(
383         np.random.default_rng(SEED + 303), means, cfg.bandit_horizon, cfg.bandit_runs
384     )
385
386     time = np.arange(cfg.bandit_horizon + 1)
387     fig, ax = plt.subplots(figsize=(7.25, 4.55))
388     ax.plot(time, ucb_curve, linewidth=1.8, label="UCB1")
389     ax.plot(time, kl_curve, linewidth=2.0, label="KL-UCB")
390     ax.plot(time, ts_curve, linewidth=1.8, label="Thompson sampling")
391     ax.set_xlabel("round $t$")
392     ax.set_ylabel("mean cumulative pseudo-regret")
393     ax.set_title("A Bernoulli-aware confidence bound spends evidence more carefully")
394     ax.legend(frameon=False)
395     fig.tight_layout()
396     fig.savefig(OUT / "kl_ucb_regret.pdf", bbox_inches="tight")
397     fig.savefig(OUT / "kl_ucb_regret.png", dpi=220, bbox_inches="tight")
398     plt.close(fig)
399
400     labels = [f"arm {i + 1}\n$\mu$={m:.2f}$" for i, m in enumerate(means)]
401     x = np.arange(len(means))
402     width = 0.25
403     fig, ax = plt.subplots(figsize=(7.25, 4.5))
404     ax.bar(x - width, ucb_counts, width, label="UCB1")
405     ax.bar(x, kl_counts, width, label="KL-UCB")
406     ax.bar(x + width, ts_counts, width, label="Thompson sampling")
407     ax.set_xticks(x, labels)
408     ax.set_ylabel("mean number of pulls")
409     ax.set_title(fr"Where the algorithms spent {cfg.bandit_horizon:,} observations")
410     ax.legend(frameon=False)
411     fig.tight_layout()
412     fig.savefig(OUT / "kl_ucb_pull_counts.pdf", bbox_inches="tight")
413     fig.savefig(OUT / "kl_ucb_pull_counts.png", dpi=220, bbox_inches="tight")
414     plt.close(fig)
415
416     rows = []
417     for name, curve, counts in (
418         ("UCB1", ucb_curve, ucb_counts),
419         ("KL-UCB", kl_curve, kl_counts),
420         ("Thompson sampling", ts_curve, ts_counts),
421     ):
422         rows.append(
423             {
424                 "experiment": "bandit",
425                 "quantity": f"final regret: {name}",

```

```

426         "value": float(curve[-1]),
427     }
428 )
429 for arm, count in enumerate(counts, start=1):
430     rows.append(
431         {
432             "experiment": "bandit",
433             "quantity": f"mean pulls arm {arm}: {name}",
434             "value": float(count),
435         }
436     )
437 return pd.DataFrame(rows)
438
439
440 def main() -> None:
441     cfg = Config()
442     rng = np.random.default_rng(SEED)
443
444     frames = [
445         plot_bernoulli_kl_geometry(),
446         plot_evidence_accumulation(rng, cfg),
447         plot_tail_bounds(cfg),
448         plot_confidence_radii(cfg),
449         plot_bandit_experiment(cfg),
450     ]
451     results = pd.concat(frames, ignore_index=True)
452     results.to_csv(OUT / "kl_divergence_results.csv", index=False)
453
454     print(results.to_string(index=False))
455
456
457 if __name__ == "__main__":
458     main()

```

## Bibliography

- [1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. 2nd ed. Wiley, 2006.
- [2] Aurélien Garivier and Olivier Cappé. “The KL-UCB Algorithm for Bounded Stochastic Bandits and Beyond”. In: *Proceedings of the 24th Annual Conference on Learning Theory*. 2011.
- [3] Z. Ghahramani. “Probabilistic machine learning and artificial intelligence”. In: *Nature* 521 (2015), pp. 452–459.
- [4] T. Jin et al. *Finite-time regret of Thompson sampling algorithms for exponential family multi-armed bandits*. arXiv:2206.03520. 2022.
- [5] Emilie Kaufmann, Olivier Cappé, and Aurélien Garivier. “On the Complexity of Best-Arm Identification in Multi-Armed Bandit Models”. In: *Journal of Machine Learning Research* 17.1 (2016), pp. 1–42.
- [6] Johannes Kirschner and Andreas Krause. “Information Directed Sampling and Bandits with Heteroscedastic Noise”. In: *Proceedings of the 31st Conference on Learning Theory*. 2018.
- [7] Solomon Kullback and Richard A. Leibler. “On Information and Sufficiency”. In: *The Annals of Mathematical Statistics* 22.1 (1951), pp. 79–86.
- [8] T. L. Lai and H. Robbins. “Asymptotically efficient adaptive allocation rules”. In: *Advances in Applied Mathematics* 6.1 (1985), pp. 4–22.
- [9] T. Lattimore and C. Szepesvari. *Bandit Algorithms*. Cambridge University Press, 2020.
- [10] David J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.
- [11] Binxin Ru et al. *Fast Information-Theoretic Bayesian Optimisation*. arXiv:1711.00673. 2018.
- [12] Daniel Russo and Benjamin Van Roy. “An Information-Theoretic Analysis of Thompson Sampling”. In: *Journal of Machine Learning Research* 17.68 (2016), pp. 1–30.
- [13] N. Srinivas et al. “Gaussian process optimization in the bandit setting: No regret and experimental design”. In: *International Conference on Machine Learning*. 2010.
- [14] J. Yang, V. Y. F. Tan, and T. Jin. *Best arm identification with minimal regret*. arXiv:2409.18909. 2024.